Bihar Engineering University, Patna

B.Tech 4th Semester Examination, 2025

Course: B.Tech | *Time: 03 Hours*

Code: 101405 | *Subject: Introduction to Solid Mechanics* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.

*(ii) There are **NINE** questions in this paper.*

*(iii) Attempt **FIVE** questions in all.*

(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) For a simply supported beam with central point load P, the maximum bending moment occurs at

(i) Support
(ii) Quarter span
(iii) Midspan
(iv) Wherever shear force is maximum

(b) A solid circular shaft transmits torque T with shear stress τ. If its diameter is doubled, the torque capacity becomes

(i) 2T
(ii) 8T
(iii) 4T
(iv) 16T

(c) Point of contraflexure in a beam is a point where

(i) Shear force is zero
(ii) Bending moment changes sign
(iii) Load is maximum
(iv) Bending moment is zero

(d) Shear stress across a rectangular beam section is maximum at

(i) Neutral axis
(ii) Bottom
(iii) Top
(iv) Both top and bottom

(e) In a thin cylindrical vessel under internal pressure, the hoop stress acts

(i) Along the length
(ii) Along the circumference
(iii) Radially outward
(iv) Radially inward

(f) A composite bar made of steel and copper is heated. The stresses induced are due to

(i) Equal modulus of elasticity
(ii) Difference in coefficients of thermal expansion
(iii) Equal thermal strain
(iv) Equal Poisson's ratio

(g) For a thin cylinder under internal pressure, the longitudinal stress is

(i) Equal to hoop stress
(ii) Twice hoop stress
(iii) Half hoop stress
(iv) Zero

(h) Hooke's Law is valid only within

(i) Plastic limit
(ii) Yield point
(iii) Elastic limit
(iv) Ultimate point

(i) The ratio of lateral strain to longitudinal strain is called

(i) Elastic modulus
(ii) Bulk modulus
(iii) Poisson's ratio
(iv) Shear modulus

(j) In a 2D stress system, the planes on which the shear stress becomes zero are known as

(i) Strain planes
(ii) Principal planes
(iii) Yield planes
(iv) Critical planes

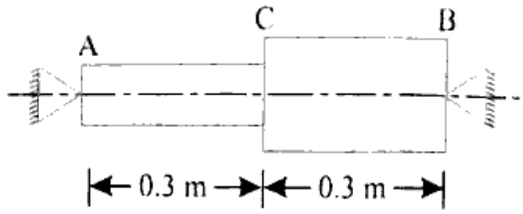
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Q.2 (a) Explain Hooke's law and the strain for elastic materials. [7]

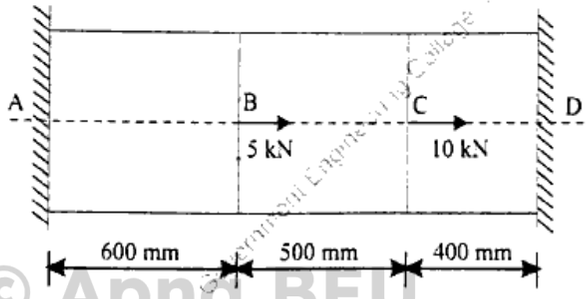
(b) Derive the bending equation $\frac{M}{I} = \frac{E}{R} = \frac{\sigma}{Y}$.

Q.3 (a) Calculate the values of the stress and strain in portions AC and CB of the steel bar shown in Figure. A close fit exists at both of the rigid supports at room tempsserature and the temperature is raised by 75°C. Take E = 200 GPa and α = 12×10^{-6} /°C for steel. Area of cross sections of AC is 400 mm² and of BC is 800 mm². [7]



(b) A C.I. flat, 300 mm long and of 30 mm × 50 mm uniform section, is acted upon by the following forces uniformly distributed over the respective cross-section; 25 kN in the direction of length (tensile); 350 kN in the direction of the width (compressive); and 200 kN in the direction of thickness (tensile). Determine the change in volume of the flat. Take $E = 140\ GN/m^2$, and $m = 4$. [7]

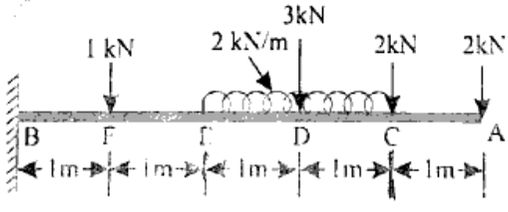
Q.4 (a) A straight uniform bar AD is clamped at both ends and loaded as shown in Figure. Initially the bar is stress free. Determine the stresses in all the three parts (AB, BC, CD) of the bar if the cross-sectional area of bar is 1000 mm^2. [7]



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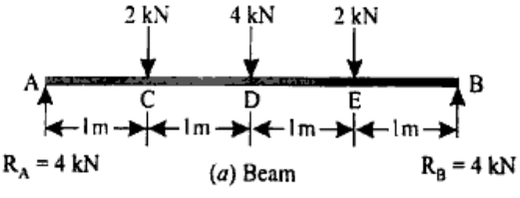
(b) Explain the different types of loads and beams with neat sketch. Also state Point of contraflexure. [7]

Q.5 (a) Draw the S.F. and B.M. diagrams for the cantilever loaded as shown in the figure. [7]



(b) Sate and explain Circumferential and longitudinal stress in thin cylinder. [7]

Q.6 (a) Draw the S.F. and B.M. diagrams for simply supported beam loaded as shown in Figure. [7]

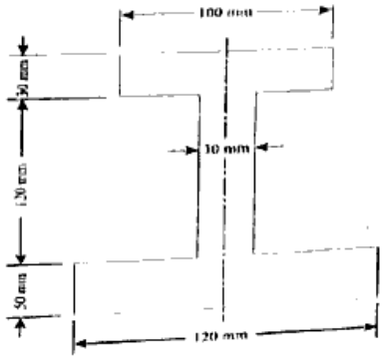


(b) Sate and explain Circumferential and longitudinal stress in thin cylinder. [7]

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Q.7 (a) Derive and state the comparison of Hollow and solid shaft by strength. [7]

(b) A beam simply supported at ends and having cross-section as shown in Figure is loaded with a U.D.L., over whole of its span. If the beam is 8 m long, find the U.D.L. if maximum permissible bending stress in tension is limited to 30 MN/m^2 and in compression to 45 MN/m^2. What are the actual maximum bending stresses set up in the section? [7]



Q.8 (a) Derive the expression of S.F and B.M. for a simply supported beam carrying a concentrated load at mid span. Also draw S.F. & B.M. diagram. [7]

(b) A hollow shaft of diameter ratio 3/8 is required to transmit 600 kW at 110 r.p.m., the maximum torque being 20% greater than the mean. The shear stress is not to exceed 63 MN m^2 and the twist in a length of 3 m not to exceed 1.4 degrees. Calculate the maximum external diameter satisfying these conditions. Take: C = 84 GN/m^2

[7 x 2 = 14]

Q.9 Write short note on *any two* of the following:

(a) Stress- Strain curve for mild steel.

(b) St. Venant's Principle.

(c) Moment area Method.

(d) Elastic Moduli.

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